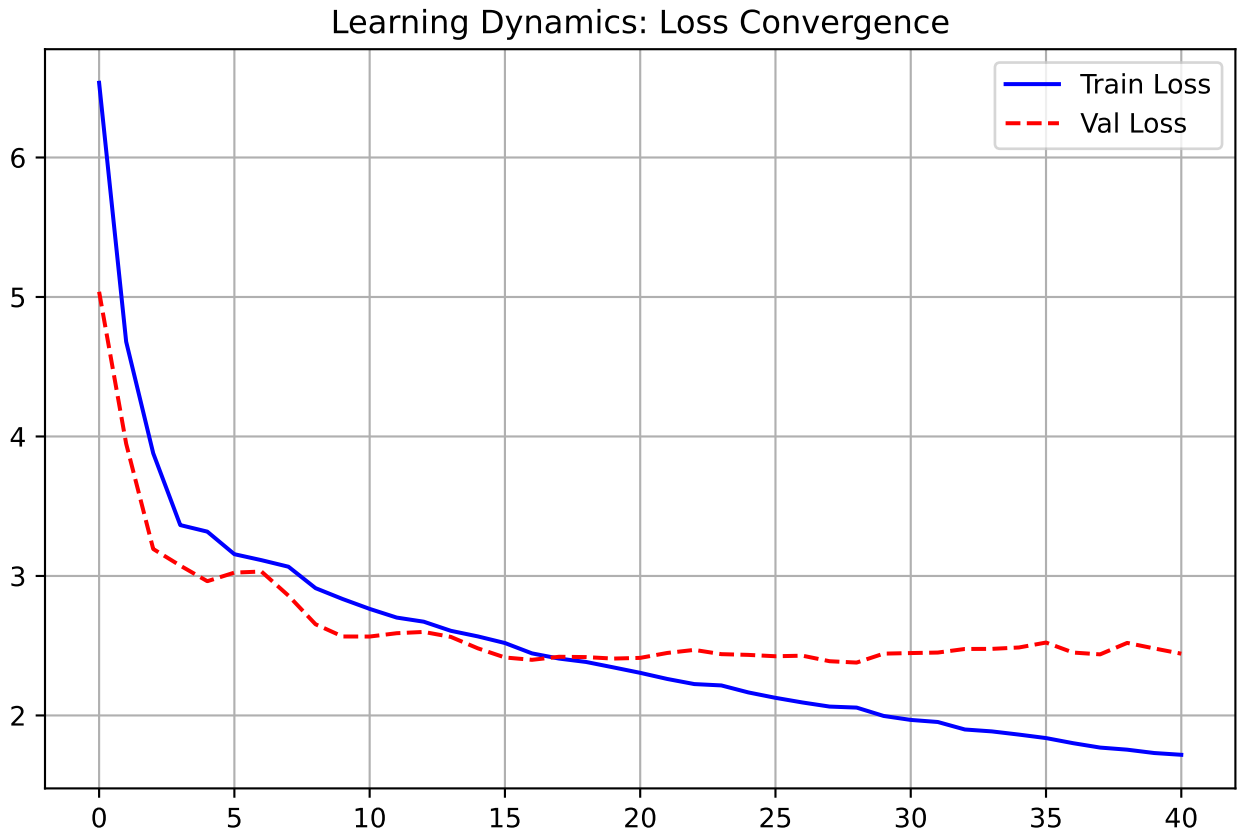
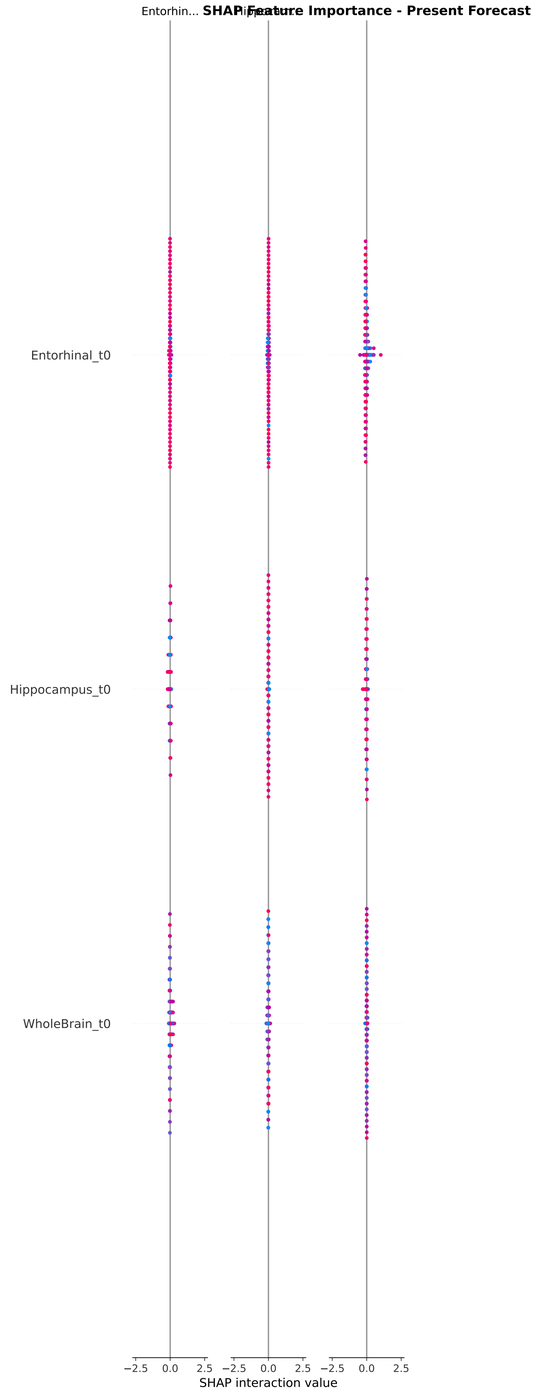
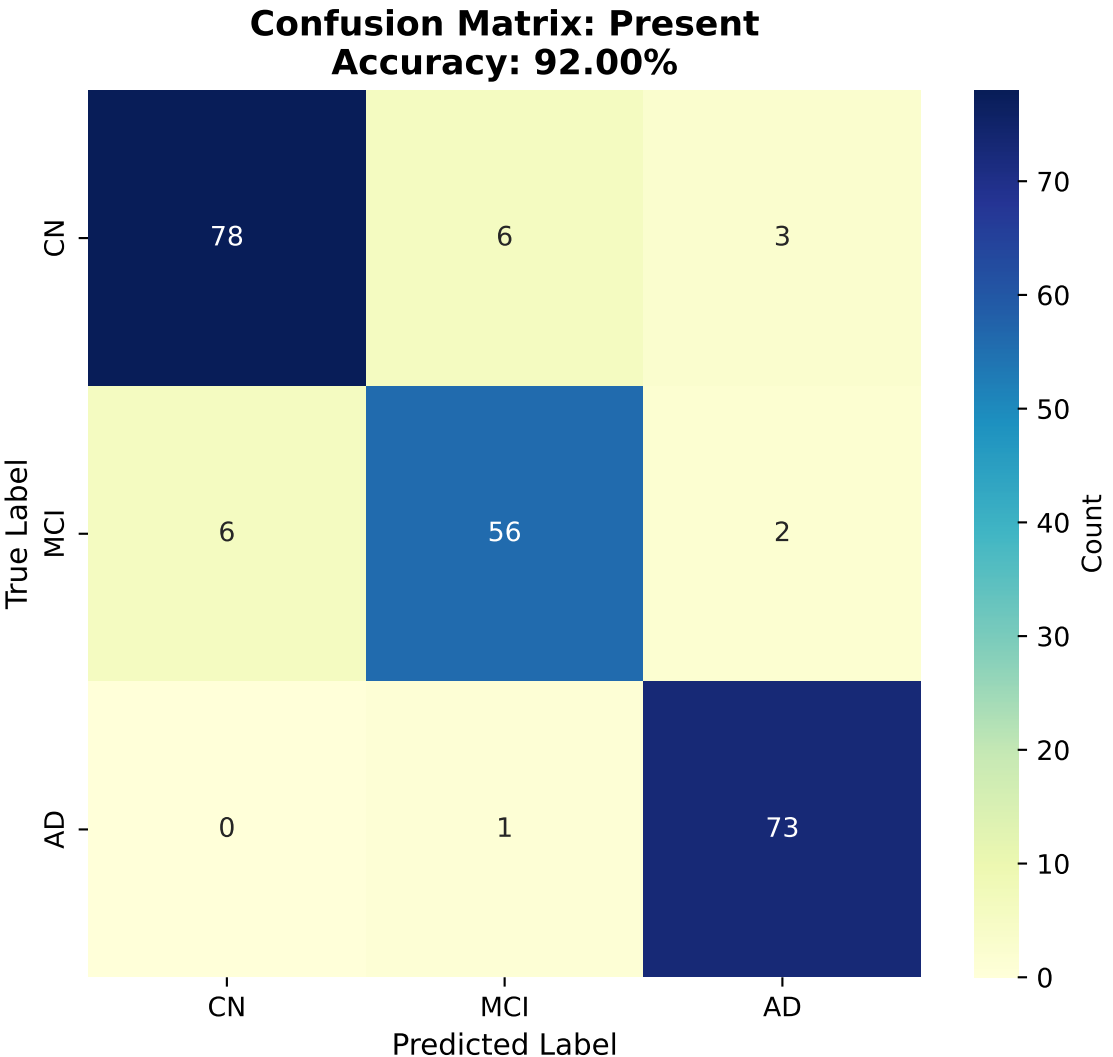
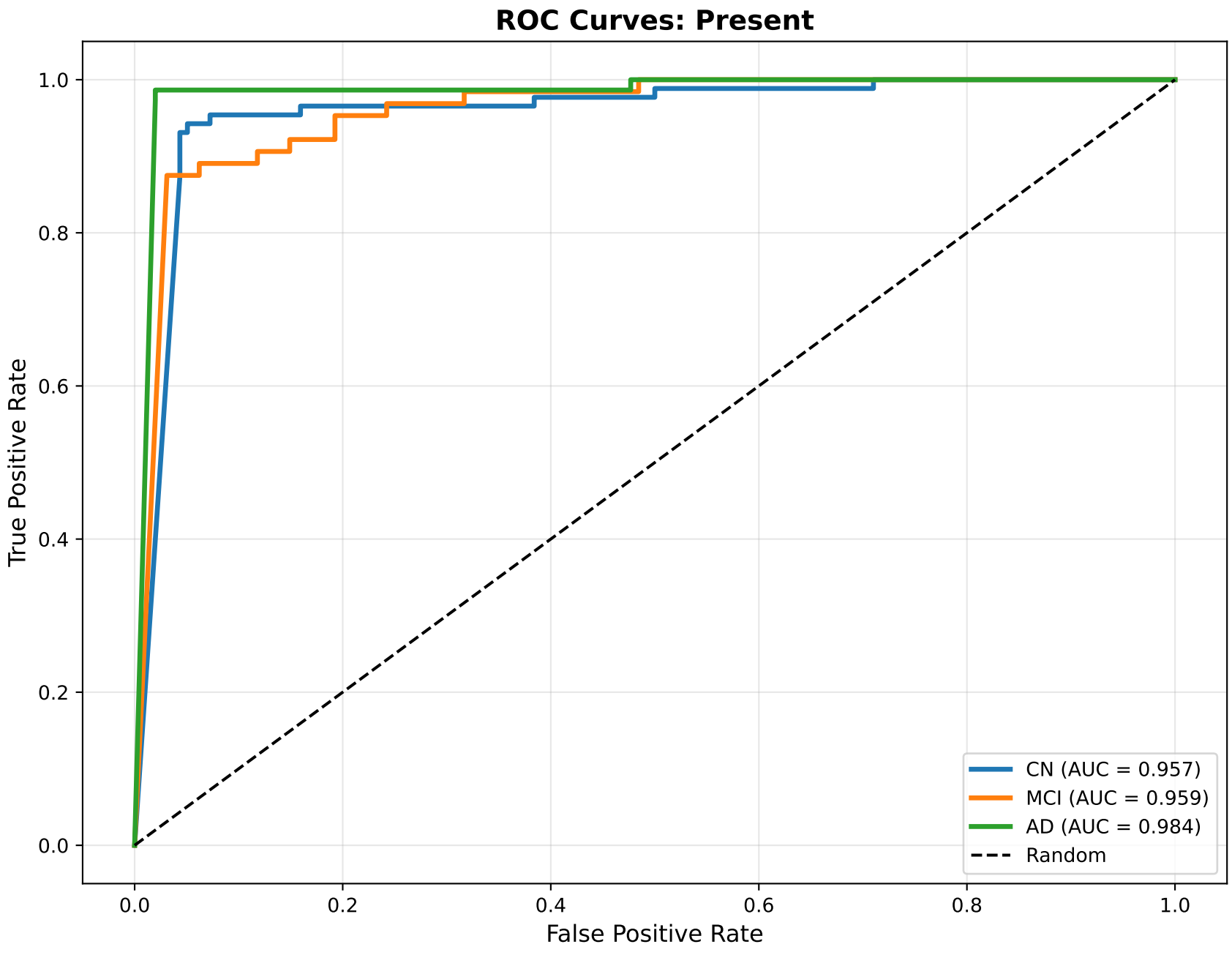
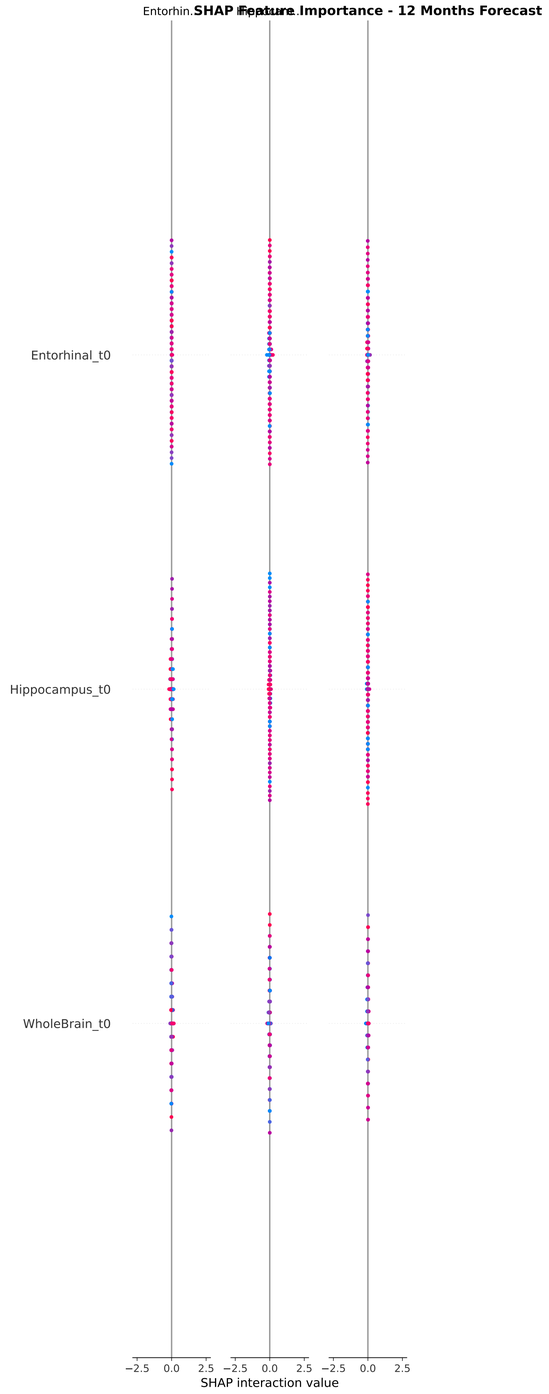
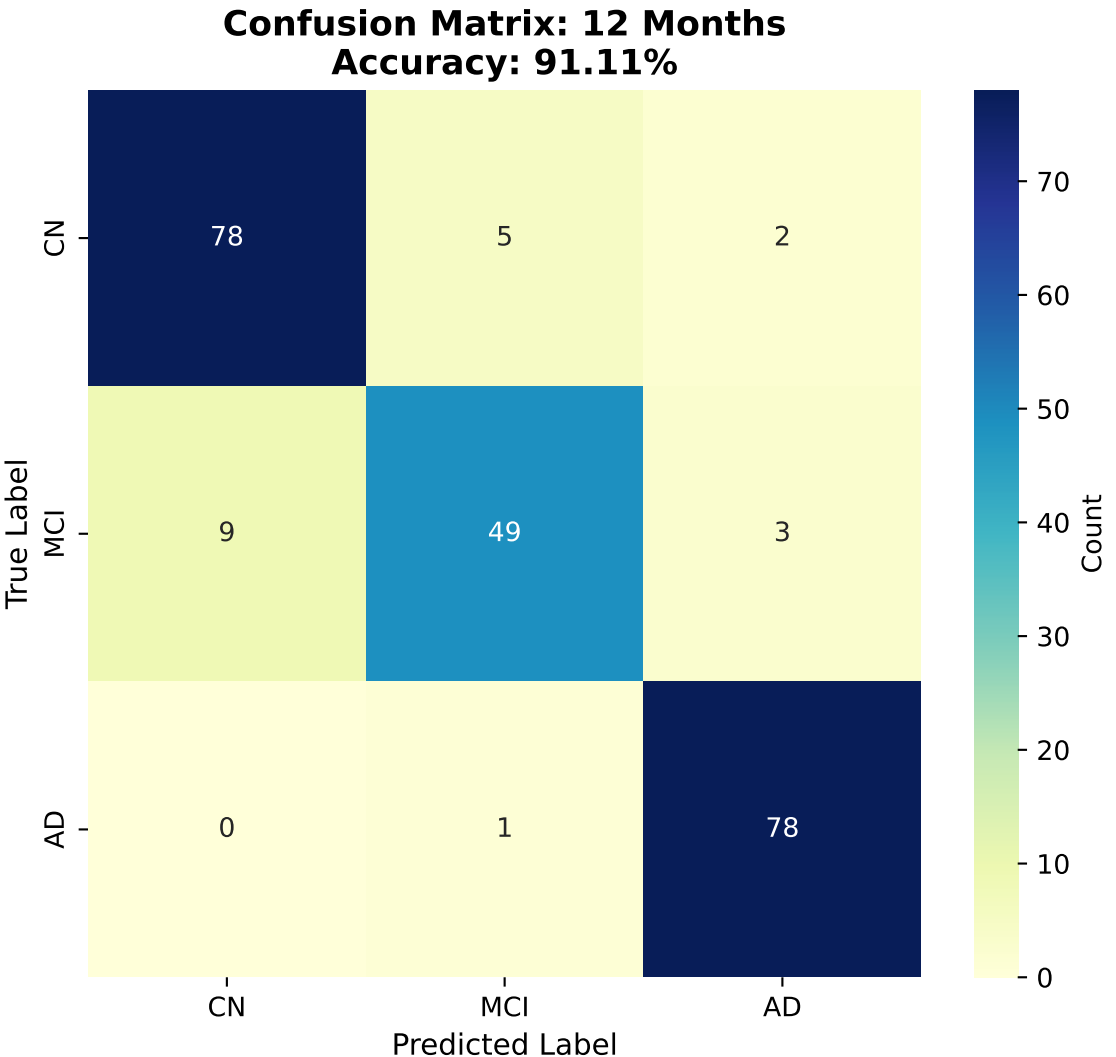

Learning Dynamics: Loss Convergence
Train Loss
Val Loss
6
5
4
3
2
0
5
10
15
20
25
30
35
40

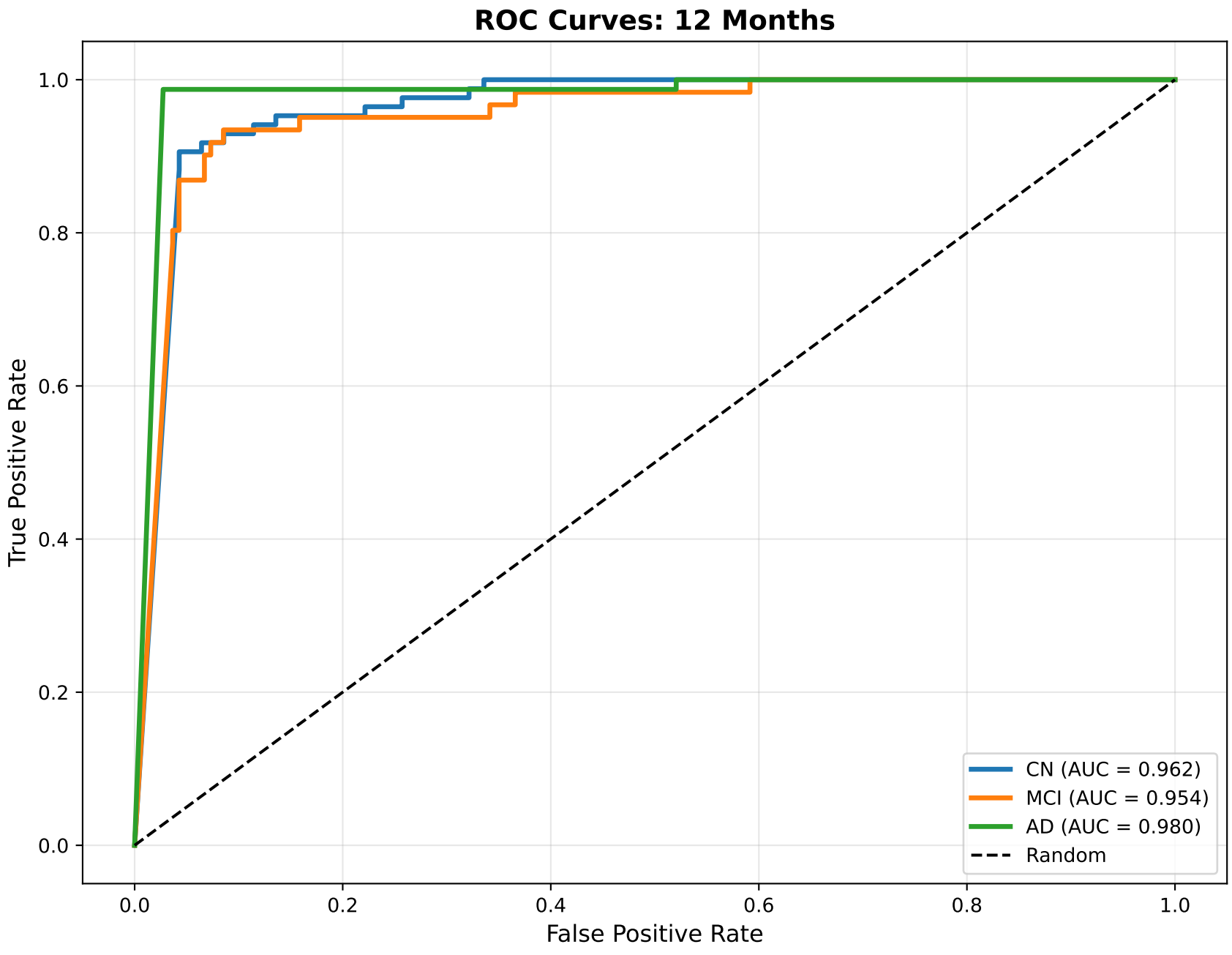

SHAP Feature Importance - Present Forecast
Entorhin...
Hippocampus
Entorhinal_t0
Hippocampus_t0
WholeBrain_t0
−2.5
0.0
2.5
SHAP interaction value

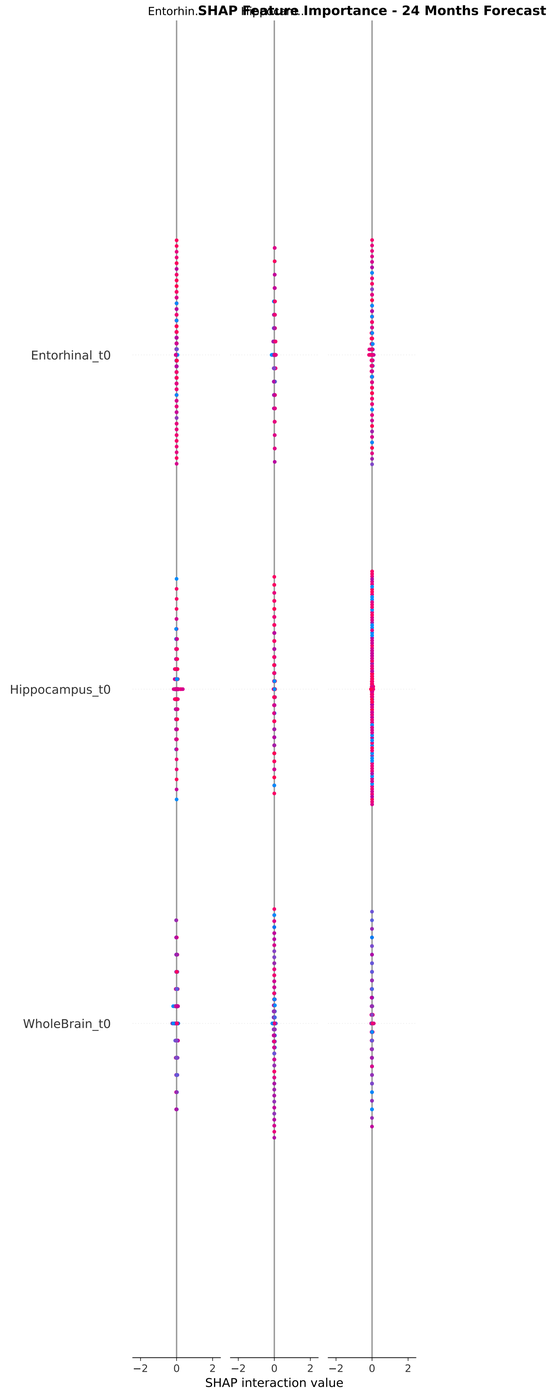

Confusion Matrix: Present
Accuracy: 92.00%
True Label
CN
MCI
AD
78
6
3
6
56
2
0
1
73
CN
MCI
AD
Predicted Label
Count
70
60
50
40
30
20
10
0

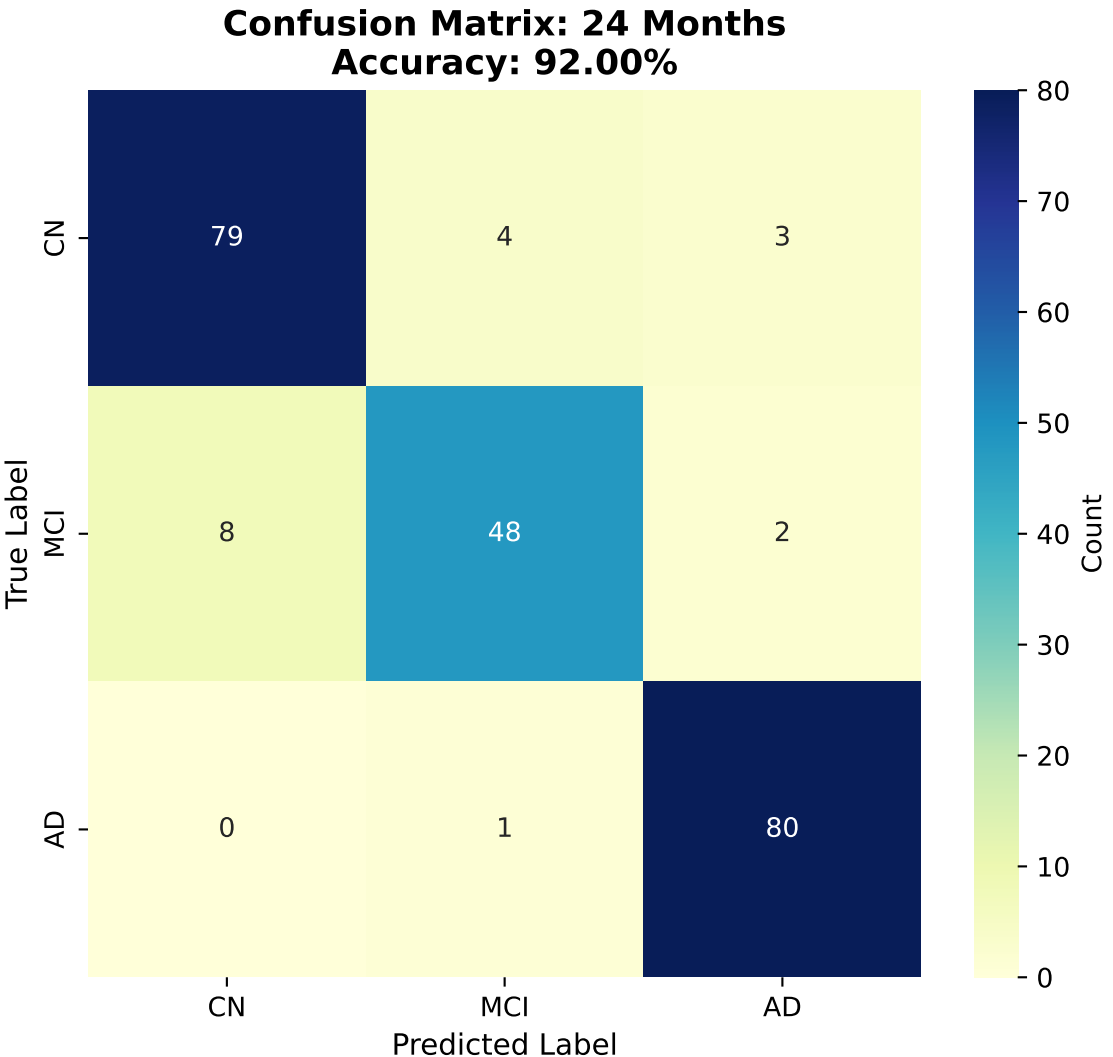

ROC Curves: Present
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
CN (AUC = 0.957)
MCI (AUC = 0.959)
AD (AUC = 0.984)
Random

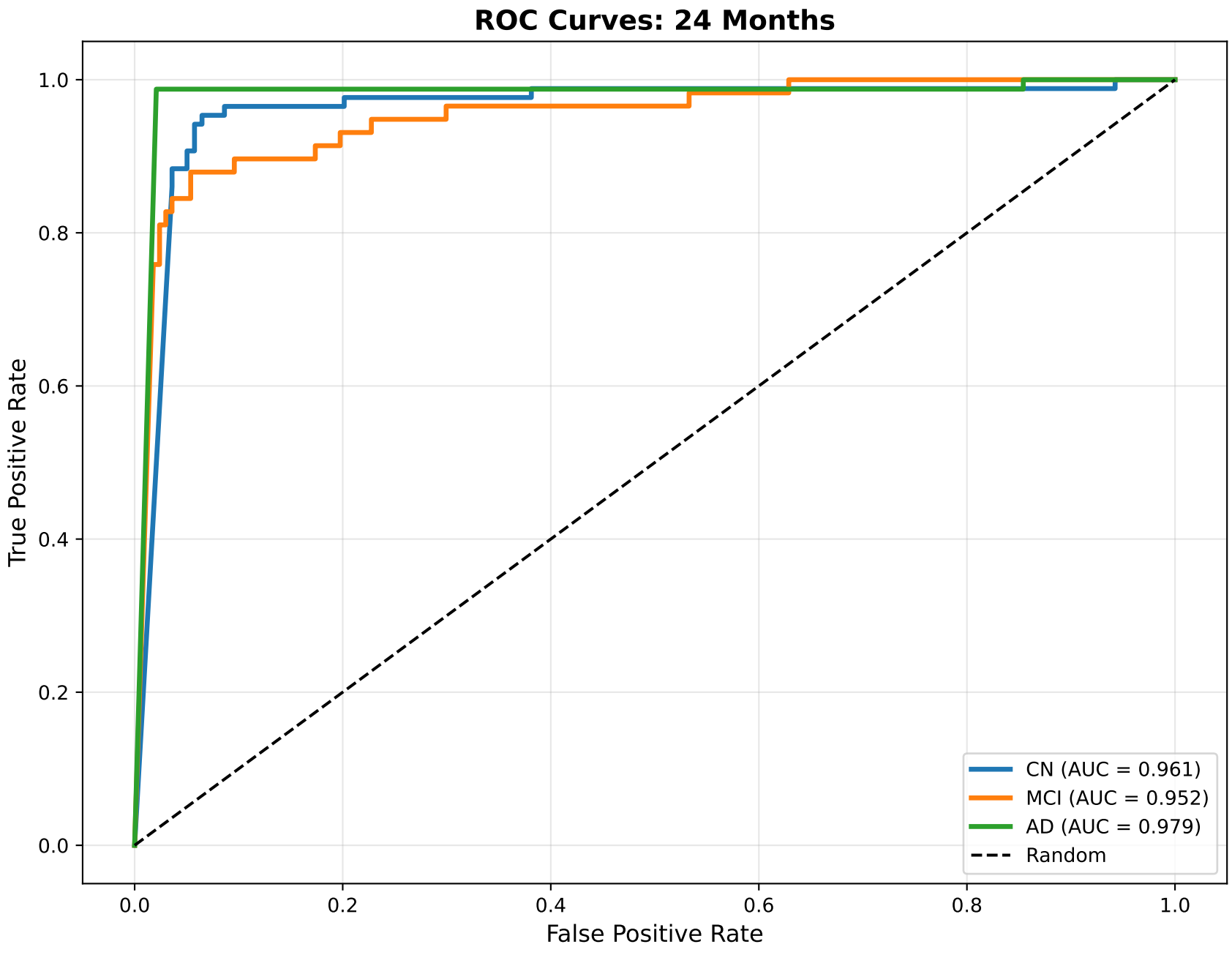

SHAP Feature Importance - 12 Months Forecast
Entorhin.
Hippocam.
Entorhinal_t0
Hippocampus_t0
WholeBrain_t0
−2.5
0.0
2.5
SHAP interaction value

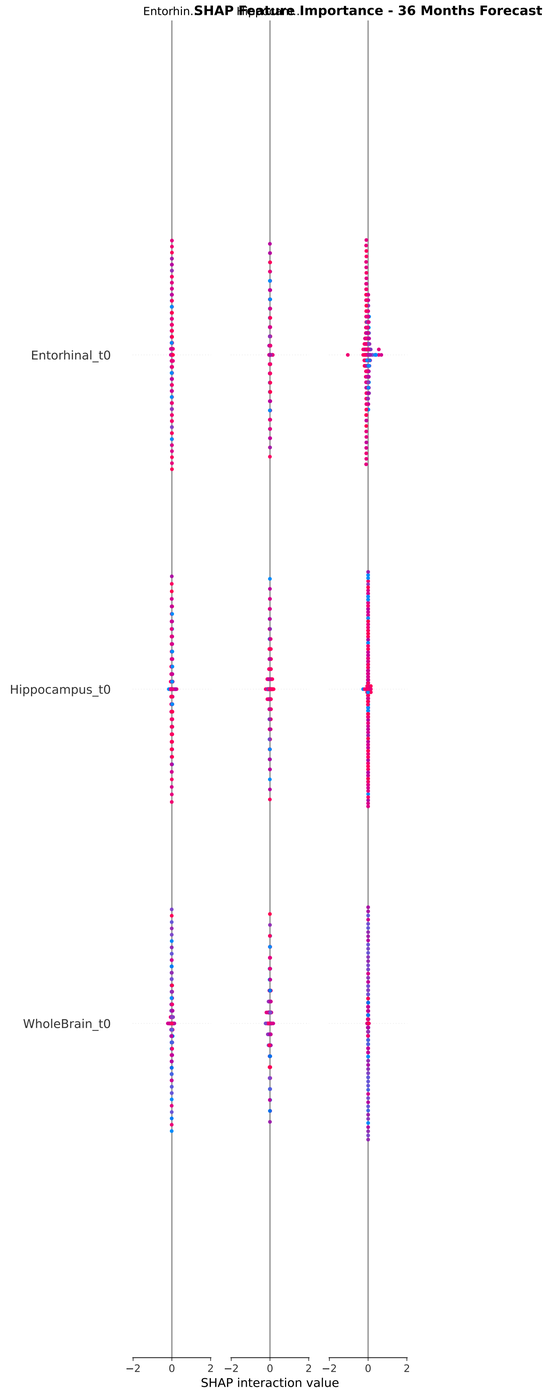

Confusion Matrix: 12 Months
Accuracy: 91.11%
True Label
CN
MCI
AD
78
5
2
9
49
3
0
1
78
CN
MCI
AD
Predicted Label
Count
70
60
50
40
30
20
10
0

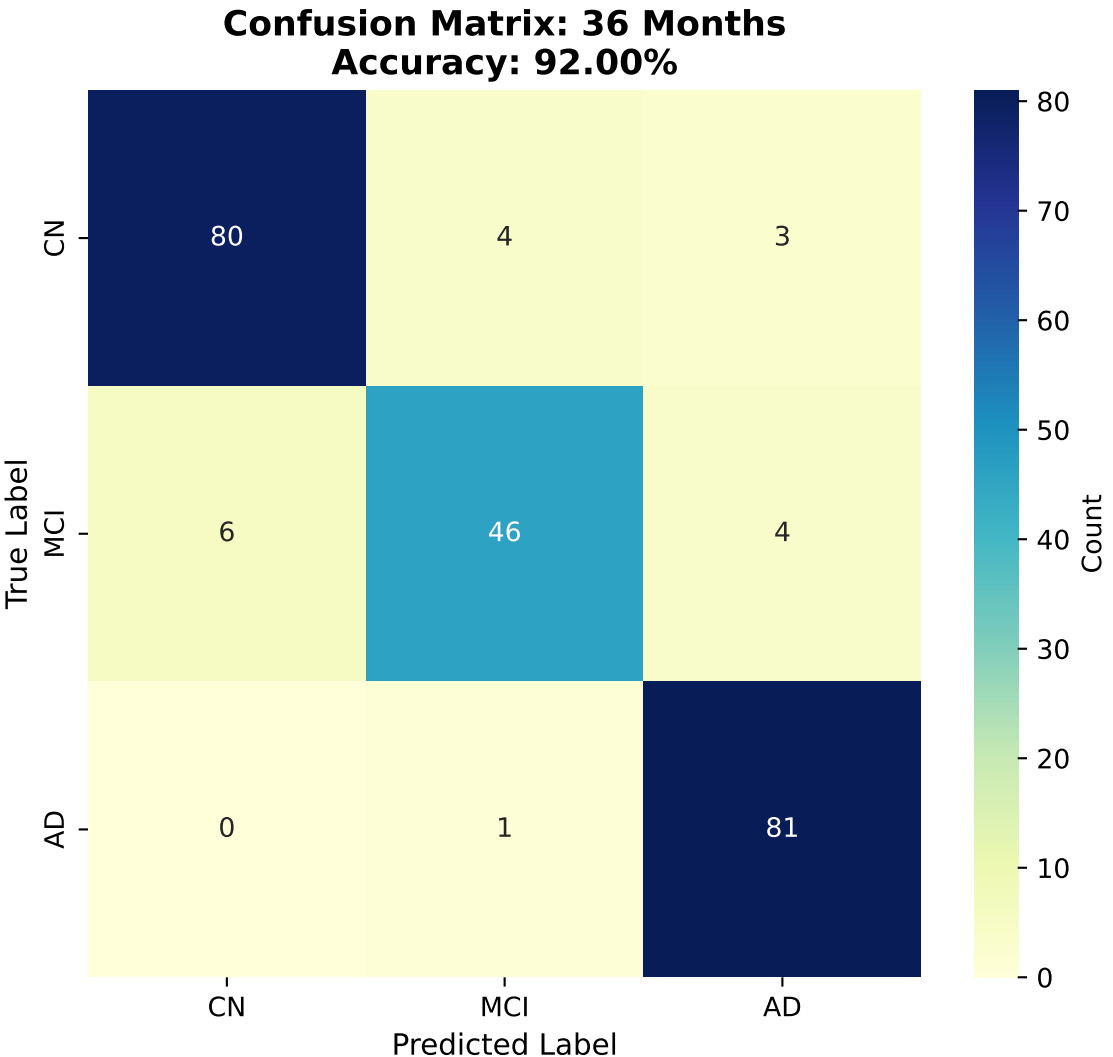

ROC Curves: 12 Months
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
CN (AUC = 0.962)
MCI (AUC = 0.954)
AD (AUC = 0.980)
Random

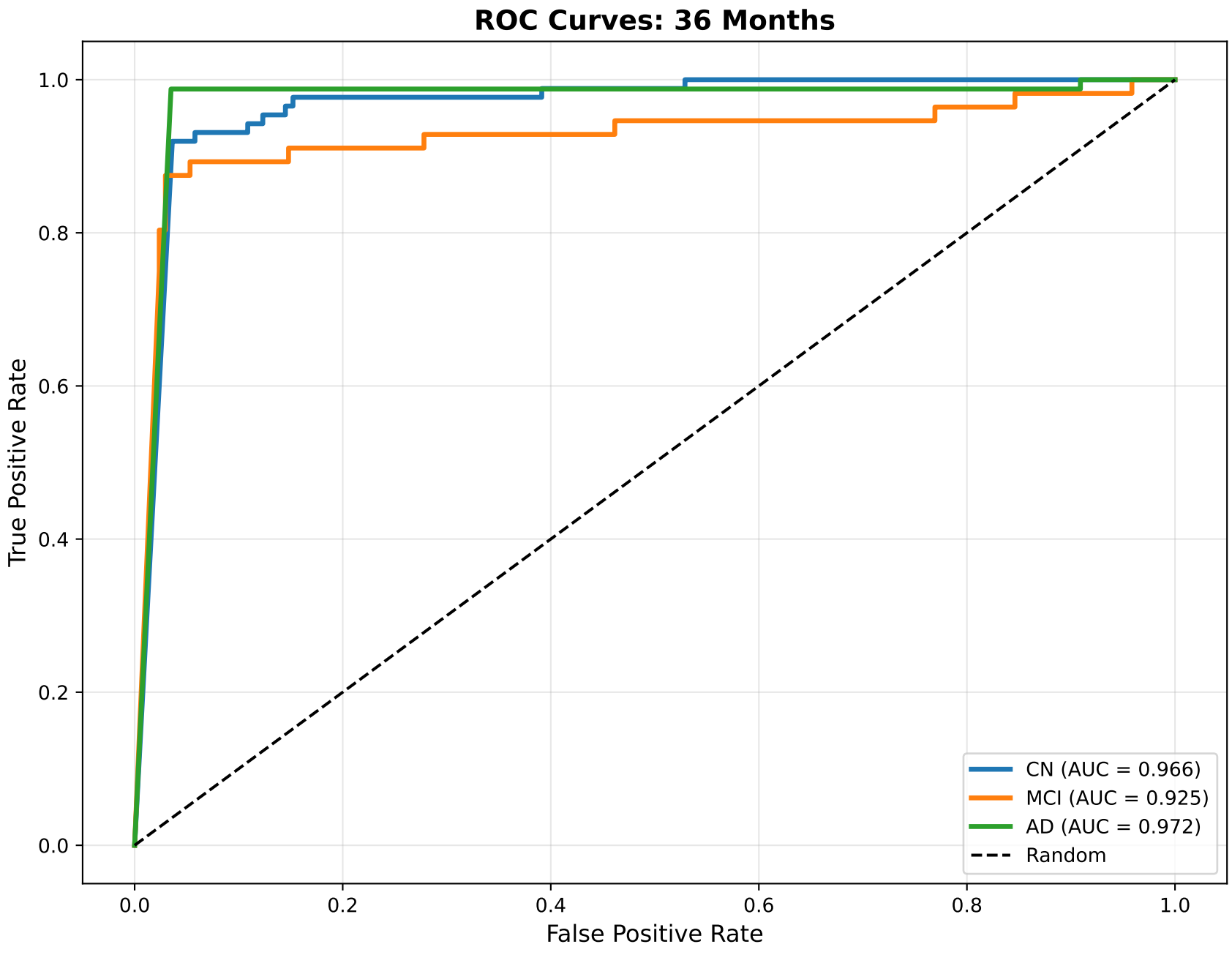

SHAP Feature Importance - 24 Months Forecast
Entorhin...
Entorhinal_t0
Hippocampus_t0
WholeBrain_t0
−2
0
2
SHAP interaction value

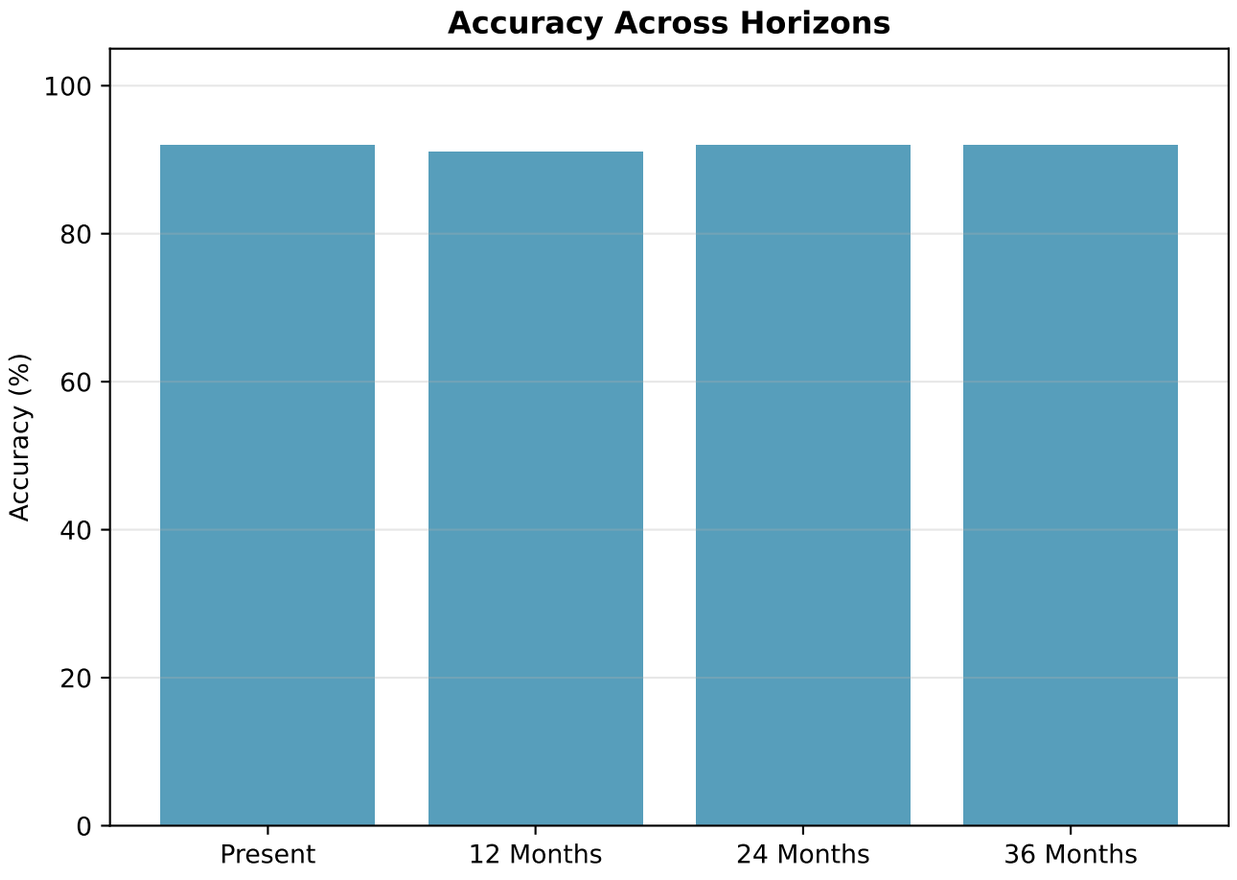

Confusion Matrix: 24 Months
Accuracy: 92.00%
True Label
CN
MCI
AD
79
4
3
8
48
2
0
1
80
CN
MCI
AD
Predicted Label
Count
80
70
60
50
40
30
20
10
0

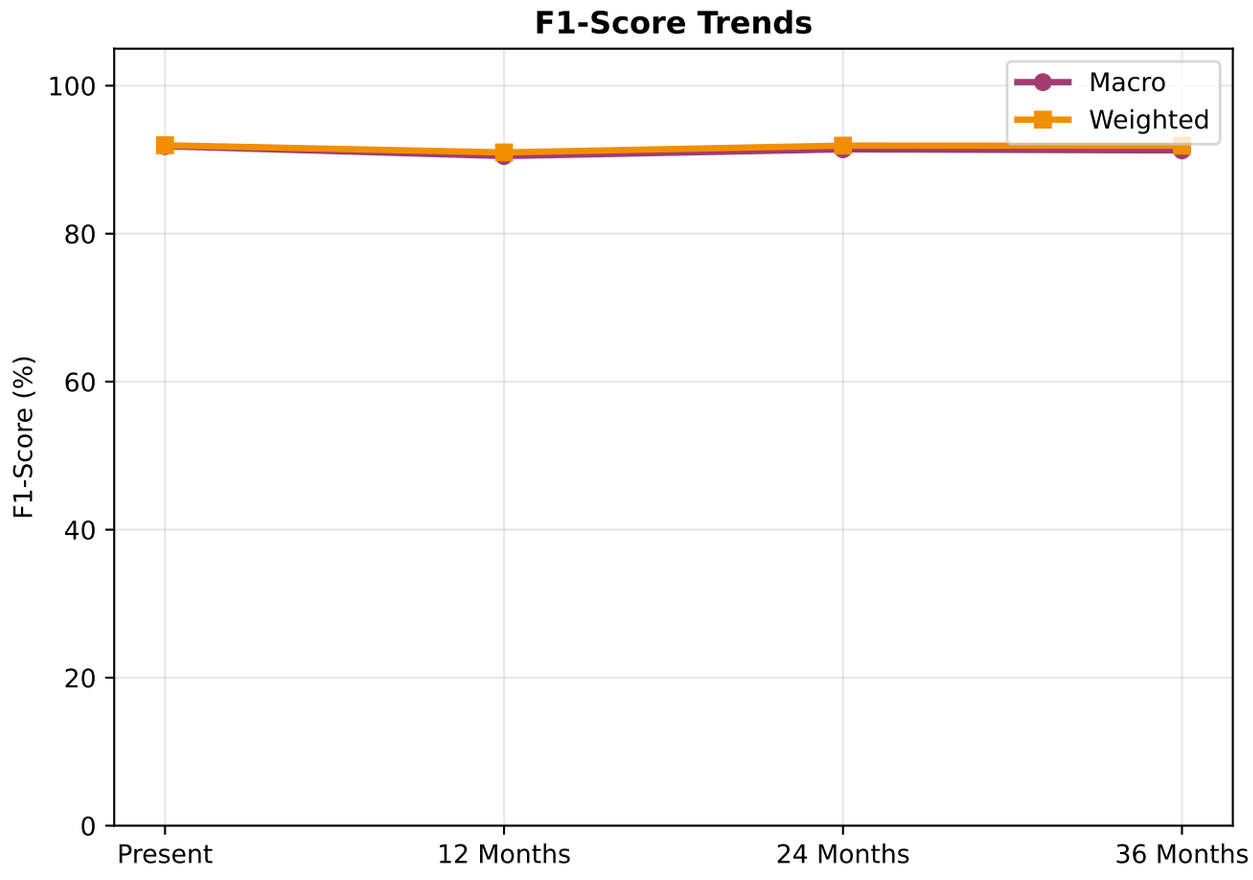

ROC Curves: 24 Months
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
CN (AUC = 0.961)
MCI (AUC = 0.952)
AD (AUC = 0.979)
Random

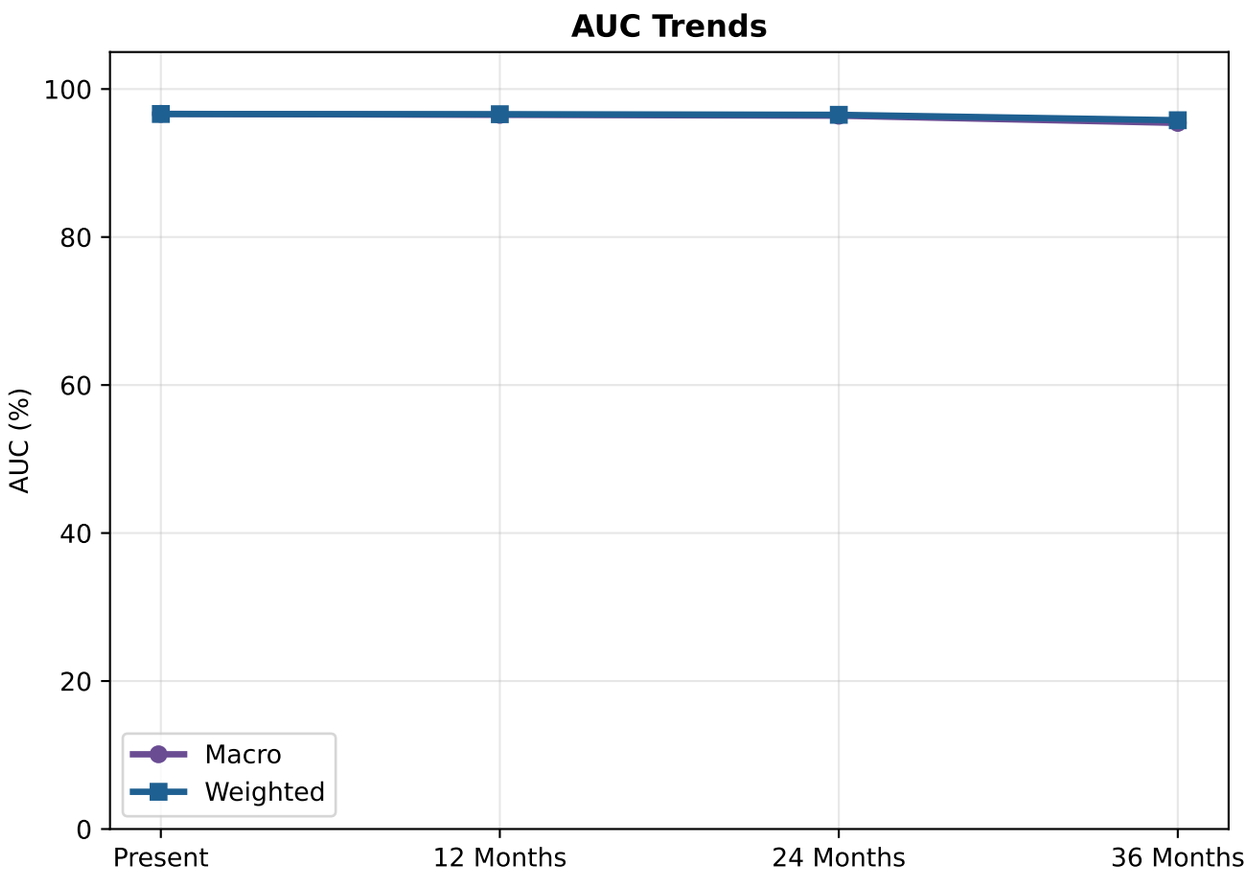

SHAP Feature Importance - 36 Months Forecast
Entorhin.
Hippocam.
Entorhinal_t0
Hippocampus_t0
WholeBrain_t0
−2
0
2
SHAP interaction value

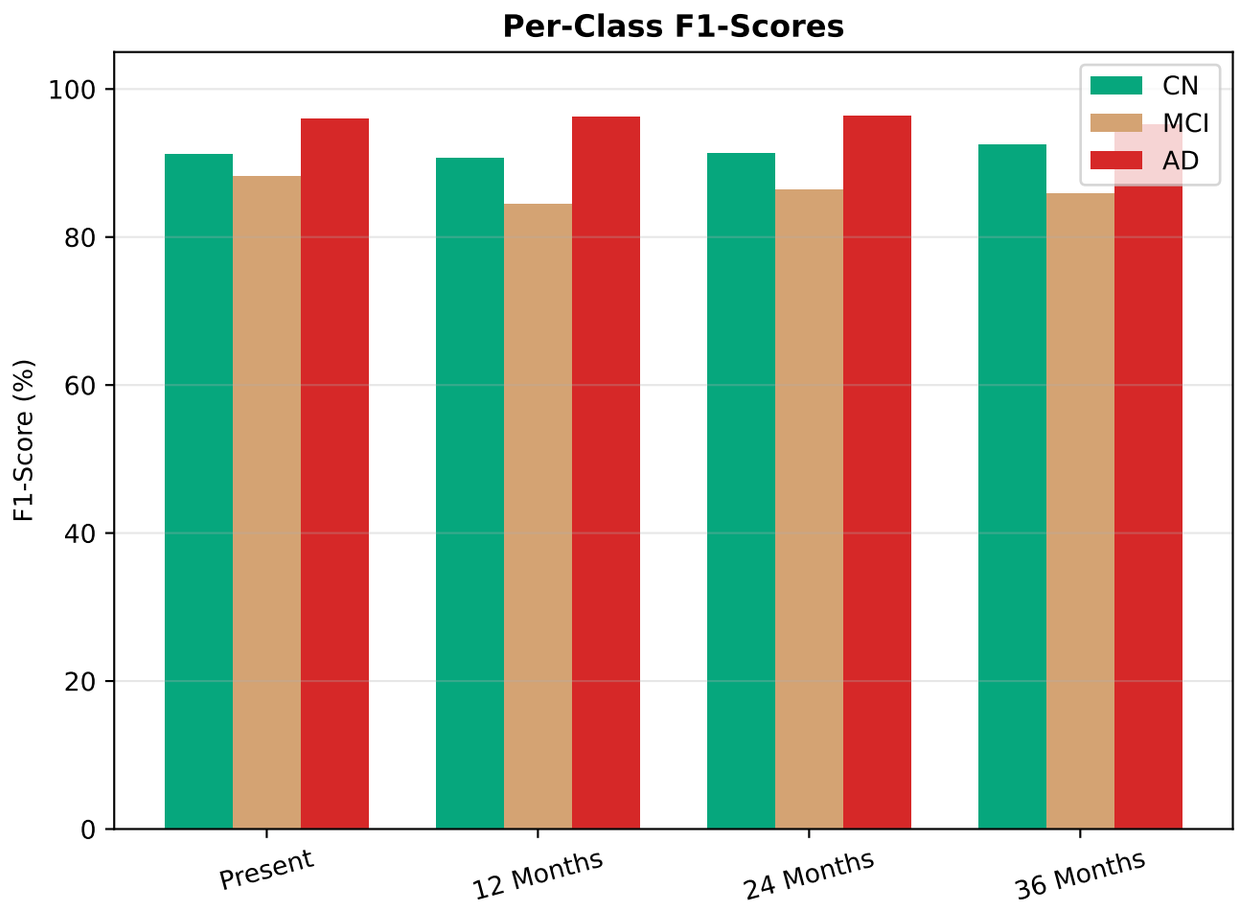

Confusion Matrix: 36 Months
Accuracy: 92.00%
True Label
CN
MCI
AD
80
4
3
6
46
4
0
1
81
CN
MCI
AD
Predicted Label
Count
80
70
60
50
40
30
20
10
0


ROC Curves: 36 Months
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
CN (AUC = 0.966)
MCI (AUC = 0.925)
AD (AUC = 0.972)
Random


Accuracy Across Horizons
Accuracy (%)
0
20
40
60
80
100
Present
12 Months
24 Months
36 Months


F1-Score Trends
F1-Score (%)
0
20
40
60
80
100
Macro
Weighted
Present
12 Months
24 Months
36 Months


AUC Trends
AUC (%)
0
20
40
60
80
100
Macro
Weighted
Present
12 Months
24 Months
36 Months


Per-Class F1-Scores
F1-Score (%)
0
20
40
60
80
100
CN
MCI
AD
Present
12 Months
24 Months
36 Months